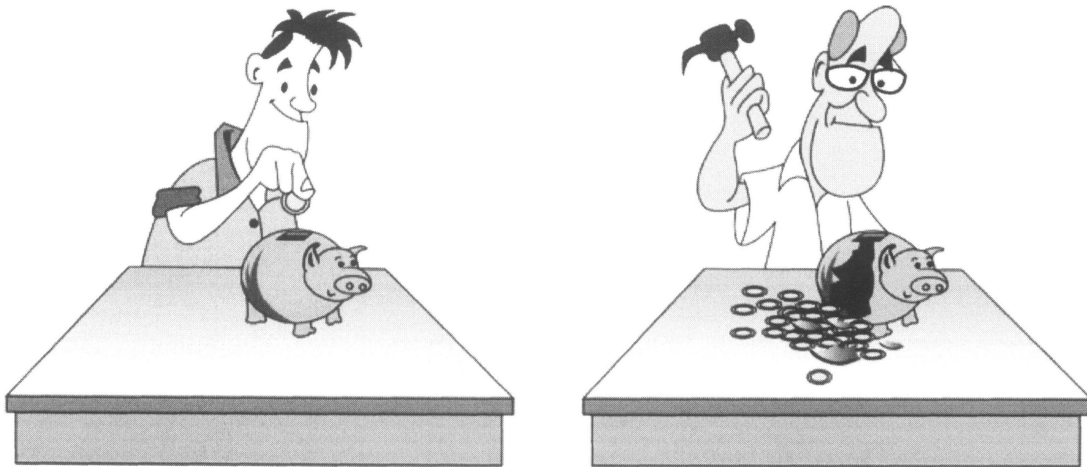


Chapter 4.

Finance II: Drawing down the investment.

Superannuation.

Retirees in Australia may, if they satisfy certain conditions, qualify for the government age pension. The amount they receive depends on what other income and assets they have. However the total of these pension payments form a considerable expenditure for the government and, with increasing life expectancy, an expenditure that could get bigger and bigger in years to come. For this reason the government encourages people to put money into savings during their years of employment so that they will have their own money to support themselves in retirement and hence only need a part pension, or perhaps even no pension payment at all, from the government. Saving to have money available in retirement is called **superannuation**. During a person's working years they **accumulate funds** in their superannuation account and then, when they retire from work, they **draw down those funds** as an income they can live on, supplemented by the government pension if the requirements of an assets and income test are met.



How much does a person need to have saved in superannuation to live comfortably in retirement?

What sort of annual income will a retired couple require?

What does the internet suggest as answers to these questions?

Investigate.

When a person is employed their employer must pay a certain percentage of their wage into a superannuation account. Thus during the employment years a person's superannuation account is in its *accumulation phase*. Then, when the employee retires, they draw money from the superannuation account as a self-funded pension, possibly supplemented by the government pension, to give them an income in retirement. The superannuation account is then in its *pension phase*. Of course in both the accumulation phase and the pension phase any funds in the account can earn interest. Thus in the pension phase interest is earned by the account and, at the same time, money is periodically withdrawn from the account.

Example 1

At retirement Susie invests the \$750 000 she has saved in her superannuation account in an account giving interest of 6% per annum compounded annually from which she will withdraw \$60 000 after one year, and after each year thereafter. How much will be left in the account immediately after the tenth withdrawal (nearest dollar)?

If she wants the account to last her at least 35 years, withdrawing a constant amount each year, what should this constant amount be (rounded down to a multiple of \$100)?

Initial amount invested: \$750 000

Account balance after 1 year: $\$750\,000 \times 1.06 - \$60\,000$

Account balance after 2 years: $(\$750\,000 \times 1.06 - \$60\,000) \times 1.06 - \$60\,000$

We have $T_{n+1} = T_n \times 1.06 - \$60\,000$,

and $T_0 = \$750\,000$

Using the recursion capability of some calculators, a spreadsheet or the inbuilt financial programs of some calculators we find that:

Immediately after the tenth withdrawal there will be \$552 288 in the account, to the nearest dollar.

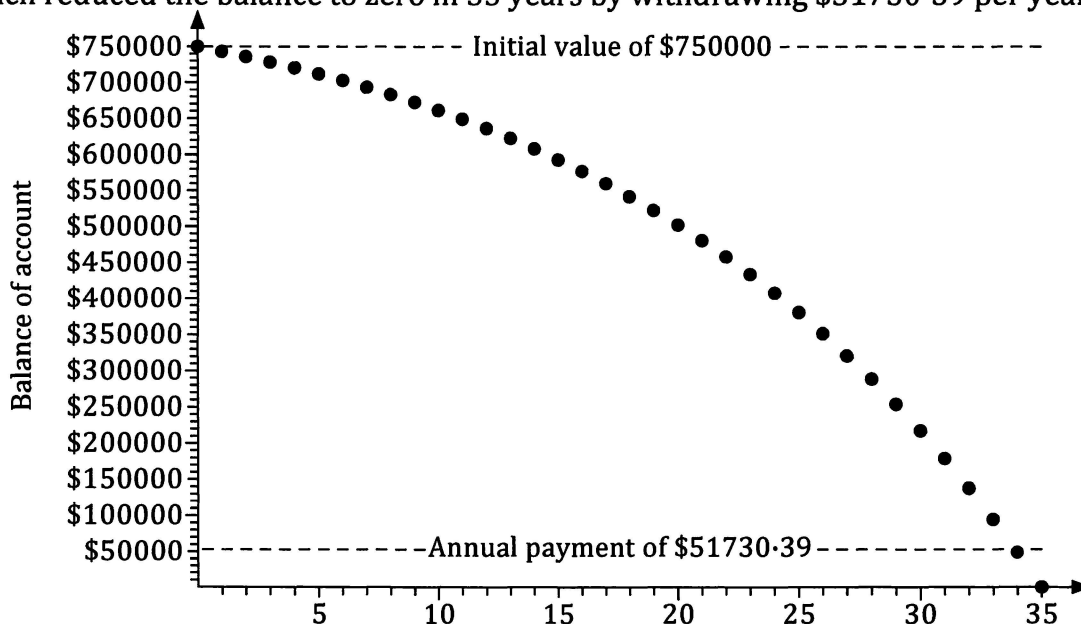
Compound Interest	
N	35
I(%)	6
PV	-750000
Pmt	51730.39423
FV	0
PpY	1
CpY	1

☒	$a_{n+1} = 1.06 \cdot a_n - 60000$
	$a_0 = 750000$
☐	$b_{n+1} = \square$
	$b_0 = 0$
☐	$c_{n+1} = \square$
	$c_0 = 0$

n	a_n
6	6.5E+5
7	6.2E+5
8	6.0E+5
9	5.8E+5
10	5.5E+5
552288.075864291	

Varying the withdrawal amount to reduce the balance to zero in 35 years, or using a calculator with an inbuilt financial capability, we can determine that Susie can withdraw up to \$51 700 each year (rounded down to a multiple of \$100) for the account to last at least 35 years.

The graph below shows the declining balance in the account of the previous example which reduced the balance to zero in 35 years by withdrawing \$51730.39 per year.



Can your calculator display a graph of this type from the terms of a recursively defined sequence? Investigate.



Annuities.

In the previous example Susie managed her own *income stream* by making a deposit into an account, withdrawing a fixed amount each year to live on and allowing the balance of the account to earn interest. A compound interest investment from which regular payments are made for a fixed period of time is called an **annuity**.

Making regular payments to pay off a loan, as encountered in the previous chapter, is a bit like an annuity. The finance company invests money with us in the form of a loan and we make regular payments back. However the word annuity is usually used in the context of using a sum of money to set up a regular income.

By making a one off payment a person can enter into an annuity arrangement whereby they will receive a guaranteed fixed regular amount for an agreed period of time. Such arrangements can be very useful for generating a steady income in retirement.

Fixed term annuities make regular payments for a fixed period of time. If the person who purchased the annuity dies before the end of the fixed period of time the remaining payments, or the remaining balance, would pass to a beneficiary.

Whole life annuities make regular payments to the purchaser of the annuity for as long as they live. Under this sort of annuity the payments stop when the person dies, i.e. upon the death of the purchaser such an annuity has no residual value. If the person dies soon after the annuity commences the finance company "wins" but if the person lives well beyond normal life expectancy the person "wins". Life annuities can be seen as an insurance in case one lives a long time. Even if the person does live a long time, regular income is assured. The finance company spreads the risk of paying out for more years than expected by having annuities with a lot of people.

Indexing

In some cases the regular amount paid out from an annuity might increase over time so that the amount received keeps up with inflation and maintains its "buying power". We say that the regular payments are *indexed*, i.e. the payments change in line with the movement of some price index, eg the consumer price index, a measure of the increase in the price of a typical household selection of goods.

Example 2

A person wishes to deposit \$200000 into an account that will earn interest of 8% per annum, compounded annually, withdrawing \$25000 at the end of the first year, \$25500 at the end of the second year, \$26010 at the end of the third year, and so on with each annual withdrawal being a 2% increase on the withdrawal of the previous year.

- (a) What will be the balance of the account immediately after the fifth withdrawal?
- (b) How long will it take for the balance in the account to reduce to zero?

- (a) Initial amount invested: \$200000
 Account balance after 1 year: $\$200000 \times 1.08 - \25000 I.e. \$191000
 Account balance after 2 years: $\$191000 \times 1.08 - \25000×1.02 I.e. \$180780
 Account balance after 3 years: $\$180780 \times 1.08 - \25000×1.02^2 I.e. \$169232.40

We have $T_{n+1} = T_n \times 1.08 - \25000×1.02^n ,
 and $T_0 = \$200000$

Using a calculator able to display the terms of a recursively defined sequence, or a spreadsheet, or by working the terms through:

T_4 , the account balance after 4 years = \$156240.79 (nearest cent)
 T_5 , the account balance after 5 years = \$141679.25 (nearest cent)

Immediately after the fifth withdrawal the balance will be \$141679.25.

- (b) Continuing the sequence:
 T_6 , the account balance after 6 years = \$125411.57 (nearest cent)
 T_7 , the account balance after 7 years = \$107290.44 (nearest cent)
 T_8 , the account balance after 8 years = \$87156.53 (nearest cent)
 T_9 , the account balance after 9 years = \$64837.57 (nearest cent)
 T_{10} , the account balance after 10 years = \$40147.26 (nearest cent)
 T_{11} , the account balance after 11 years = \$12884.18 (nearest cent)

After a further year this balance will grow to $\$12884.18 \times 1.08$
 = \$13914.91

The balance will reduce to zero at the end of the 12th year when the withdrawal will be just \$13914.91.

- Note • Care needs to be taken when using on-line annuity calculators. Some annuities work on the basis that the first payout from the annuity occurs when the annuity is opened, eg as soon as a superannuation amount is rolled into pension phase, and some work on the basis that the first payout occurs at the end of the first month, or year, whatever the agreed frequency of payment is.

Perpetuities.

Suppose we invest \$5000 in an account paying 8% per annum compounded annually, and we withdraw \$400 at the end of each year.

Initial amount invested:	\$5000		
Account balance after 1 year:	$\$5000 \times 1.08 - \400	I.e. \$5000.	
Account balance after 2 years:	$\$5000 \times 1.08 - \400	I.e. \$5000.	
Account balance after 3 years:	$\$5000 \times 1.08 - \400	I.e. \$5000.	Etc.

In this case the annual withdrawal of \$400 could continue forever because each year the interest earned exactly equals the amount paid out. This sort of annuity is called a **perpetuity**, the word perpetuity meaning *lasting forever*.

Consider the case for example of someone wishing to fund an annual award of \$400 to be given to a deserving student in a school for some reason. By donating \$5000 into an account paying 8% per annum the interest produced will fund the award forever, i.e. in perpetuity, endlessly.

Exercise 4A.

1. An ex-pupil of a school wishes to invest a sum of money into an account paying 7.5% interest, compounded annually, so that every year thereafter \$600 can be made available from the account as an award to a deserving student.
Explaining your reasoning determine how much money needs to be invested to make this perpetual award possible.
2. Someone who has made their fortune from mining wishes to set up an account that will allow a research grant of \$75000 to be paid every year thereafter to fund research at a University School of Mines Department. So that the grant can be perpetual the account, which pays 8% per annum interest, compounded quarterly, needs to generate annual interest of at least \$75000.
Explaining your reasoning determine how much the initial "one-off" investment needs to be (rounded up to the next dollar).
3. What initial investment is required into an account that pays annual interest of 6%, compounded monthly, if the interest earned each year is to pay for an annual perpetual award of \$15000?

4. Let us suppose that \$ P is invested in an account that pays $R\%$ per annum, compounded annually. At the end of the first year, and every year thereafter, \$ A is withdrawn from the account and after exactly Y years the balance of the account reaches zero.

For each of the following state the option that will increase Y , other quantities remaining the same.

- (a) Increase P or decrease P .
- (b) An increase in R or a decrease in R .
- (c) An increase in A or a decrease in A .

5. At retirement Kelvin invests the \$620 000 he has in his superannuation account in an account giving interest of 5.8% per annum compounded annually from which he will withdraw \$50 000 at the end of every twelve months to live on. Using a recursive rule determine how much will be left in the account immediately after the tenth withdrawal (nearest ten dollars).

For how many years will Kelvin be able to withdraw \$50 000 per year?

For how many years would Kelvin have been able to withdraw \$50 000 per year if instead the interest rate had been 7.8% ?

6. Julie is left \$765 000 in a will and decides to invest \$700 000 of this money in an account paying 5.4% interest compounded annually. Julie plans to retire early and use this money to live on by withdrawing \$50 000 at the end of the first year and each year thereafter until the account balance is exhausted. Use a recursive rule to determine for how many years will she be able to withdraw \$50 000 from this account.

If instead Julie withdrew \$45 000 at the end of each year for how many years could she do this?

Can you foresee any problems with the plan to withdraw a constant amount each year for a long period of time?

7. Use the financial capability of some calculators or computer programs to determine, to the nearest dollar, how much should be invested in an annuity earning 12% annual interest, compounded annually, to provide a regular annual income of \$45 000 for exactly 25 years. (The first payment is to be at the end of the first 12 months).
8. Repeat the previous question but now for an interest rate of just 6% per annum compounded annually.
9. Use the financial capability of some calculators or computer programs to determine, to the nearest dollar, how much should be invested in an annuity earning 8% annual interest, compounded monthly, to provide a regular monthly income of \$3000 for exactly 20 years. (The first payment is to be at the end of the first month).
10. Repeat the previous question but now for an interest rate of just 5% per annum compounded annually.

11. A finance company offers an annuity in which a person deposits a lump sum of \$300 000 into an account and interest is compounded annually at the rate of 7.5% per year. At the end of the first year, and every year thereafter for as long as the balance in the account allows, the annuity pays out \$30 000 with the final payment, which might be less than \$30 000, reducing the balance to zero. (In the event of death the remaining funds will be paid into the estate of the deceased.)
 - (a) Use a recurrence rule to determine how many \$30 000 payments will be made and what the final payment will be (assuming the person does not die in the meantime).
 - (b) Repeat part (a) but now determine the answer using a calculator with financial capability.

12. A finance company offers an annuity in which a person deposits a lump sum of \$500 000 into an account and interest is compounded monthly at the rate of 0.4% per month. At the end of the first month, and every month thereafter, the account pays out \$4000 with the final payment, which might be less than \$4000, reducing the balance to zero. (In the event of the death of the annuity holder remaining funds will be paid into the estate of the deceased.)
 - (a) Use a recurrence rule to determine how many \$4000 payments will be made and what the final payment will be (assuming the person does not die in the meantime).
 - (b) Repeat part (a) but now determine the answer using a calculator with financial capability or computer based financial software.

13. To the nearest dollar, how much should be invested in an annuity paying 10% annual interest, compounded annually, to provide a regular annual income of \$42000 paid at the end of each year, for exactly 25 years?
Determine the answer by
 - (a) using a recursive formula with an initial "guess" of \$400 000,
and
 - (b) using the financial capability of some calculators or computer programs.

14. Tony wins \$500 000 on the lottery and decides to use \$400 000 of it to purchase a 15 year annuity that pays a fixed amount every year for fifteen years, commencing one year after the initial purchase. The annuity adds interest to the account balance at 8% per annum, compounded annually, and deducts the regular payment at the end of each year. At the end of the 15 years the account balance is zero.
How much does Tony receive each year (to the nearest dollar)?
Determine the answer by
 - (a) using a recursive formula with an initial "guess" of \$50 000,
and
 - (b) using the financial capability of some calculators or computer programs.

15. A person wishes to deposit \$300 000 into an account that will earn interest of 8% per annum, compounded annually, withdrawing \$30 000 at the end of the first year, \$30 900 at the end of the second year, \$31 827 at the end of the third year, and so on with each annual withdrawal being a 3% increase on the withdrawal of the previous year.
 - (a) What will be the balance of the account immediately after the third withdrawal?
 - (b) How long will it take for the balance in the account to reduce to zero?

16. A person wishes to deposit \$250 000 into an account that will earn interest of 7.5% per annum, compounded annually, withdrawing \$25 000 at the end of the first year, \$26 000 at the end of the second year, \$27 040 at the end of the third year, and so on with each annual withdrawal being a 4% increase on the withdrawal of the previous year.
 - (a) What will be the balance of the account immediately after the eighth withdrawal?
 - (b) How long will it take for the balance in the account to reduce to zero?

Suppose the frequency of payments $\neq$ the frequency of compounding.

So far in this chapter, in all of the annuity situations (apart from two questions involving perpetuities) the frequency of the payments, eg annual, quarterly, monthly etc., has been the same as the frequency of the compounding periods, eg annual, quarterly, monthly etc. Let us now consider situations in which compounding occurs more frequently than the payments are made (see example 3) and situations in which payments occur more frequently than compounding (see example 4)

Example 3.

Suppose that \$250 000 is invested into an account paying 8% interest, compounded quarterly, with \$40 000 withdrawn from the account at the end of the first year and at the end of every year thereafter.

Determine the balance of the account immediately after the sixth withdrawal.

Initial amount invested:	\$250 000	
Account balance after 1 year:	$\$250\,000 \times 1.02^4 - \$40\,000$	I.e. \$230 608.04.
Account balance after 2 years:	$\$230\,608.04 \times 1.02^4 - \$40\,000$	I.e. \$209 617.56 (nearest cent)

We have $T_{n+1} = T_n \times 1.02^4 - \$40\,000$,
and $T_0 = \$250\,000$

Displaying the terms of this sequence on a calculator, as shown on the right, we see that the balance of the account immediately after the sixth withdrawal is \$106866.65 (to the nearest cent).

Alternatively the same answer could be obtained using the inbuilt financial capability of some calculators, see the typical display below.

Compound Interest	
N	6
I%	8
PV	-250000
PMT	40000
FV	106866.6549
P/Y	1
C/Y	4

☒	$a_{n+1} = 1.02^4 \cdot a_n - 40000$
	$a_0 = 250000$
☐	$b_{n+1} = \square$
	$b_0 = 0$
☐	$c_{n+1} = \square$
	$c_0 = 0$

n	a_n
2	2.1E+5
3	1.9E+5
4	1.6E+5
5	1.4E+5
6	1.1E+5
106866.654903457	

Notice that in the previous example our use of the multiplication factor 1.02^4 brought our quarterly compounding up to its equivalent effective annual compounding and thereby had the frequency of compounding matching the frequency of payment.

Now suppose that payments occur more frequently than compounding. For example payments occurring every quarter, compounding occurring annually and an annual interest rate of 8%. We can again adjust the compounding period to match the payment period by finding the quarterly multiplication factor that will produce an annual compounding of 8%. This quarterly multiplication factor is 1.019426547 because, if

$$x^4 = 1.08 \quad \text{then} \quad x = 1.019426547.$$

However, whilst we would need this equivalent multiplication factor if we were using a recursive approach, the use of a calculator with built in financial capability gives a ready alternative that avoids us having to calculate this figure, as shown in the next example.

Example 4.

Suppose that \$250 000 is invested into an account paying 8% interest, compounded annually, with \$10 000 withdrawn from the account three months later and a further \$10 000 each quarter thereafter.

Determine the balance of the account after three years.

The display shows that after 3 years (12 quarterly payments of \$10 000, i.e. 4 payments per year) an account paying 8% interest per annum, compounded annually (1 compounding per year), and initially worth \$250 000, will be worth \$181 238.775 after three years.

Compound Interest	
N	12
I%	8
PV	-250000
PMT	10000
FV	181238.775
P/Y	4
C/Y	1

Notice that using the 1.019426547 multiplication factor mentioned on the previous page to create an appropriate recursive rule gives this same answer, as shown below. However for these questions, in which the payments occur more frequently than the compounding, using the financial capability of some calculators is recommended rather than using a recurrence relation.

$a_{n+1} = 1.019426547 \cdot a_n - 10000$	
$a_0 = 250000$	
n	a_n
8	2.1E+5
9	2.0E+5
10	1.9E+5
11	1.9E+5
12	1.8E+5
181238.775301663	

Exercise 4B.

1. Use a recursive approach for this question.
Suppose that \$400 000 is invested into an account paying 6% per annum interest, compounded semi-annually, with \$35 000 withdrawn from the account at the end of the first year and at the end of every year thereafter.
Determine the balance of the account immediately after the tenth withdrawal.
2. For this question use a recursive approach (or a spreadsheet of your own creation) and make \$500 000 your "first guess" at the amount invested.
To the nearest dollar, how much needs to be invested in an account paying interest of 6% per annum, compounded quarterly, for \$45 000 to be withdrawn at the end of every year for 25 years?
3. For this question use the financial capability of some calculators.
How much needs to be invested, to the nearest cent, in an account paying interest of 9% per annum, compounded annually, for \$10 000 to be paid at the end of the first 3 months, and every three months thereafter for 15 years, at the end of which time the account has zero balance remaining?
4. For this question use the financial capability of some calculators.
How much needs to be invested, to the nearest cent, in an account paying interest of 6% per annum, compounded annually, for \$4 000 to be paid at the end of the first month, and every month thereafter for 20 years, at the end of which time the account has zero balance remaining?
5. Upon retirement John switches his superannuation fund of \$480 000 into an account paying 9% per annum compounded monthly. After three months, and every three months thereafter John wants the account to pay him \$15 000, for as long as possible.
How many payments of \$15 000 will this scheme allow John to receive and what will be the final payment that closes the account?

Do some research and write a brief report on some of the following questions.

What incentives and rules does the Government of Australia put in place to ensure / encourage people to put money into superannuation?

What is the current average balance that people have in superannuation?

Why is an ageing population a challenge for governments?

What is the retirement age in Australia? What is it in other countries?

COMPUTER SPREADSHEET OR GRAPHIC CALCULATOR ACTIVITY

Sue wins \$120 000 in a lottery. Rather than spend it all immediately she decides to use some of it, and perhaps even all of it, to allow herself to take some years off work, or at least reduce her hours somewhat.



She asks your advice regarding how long the money would last under each of five schemes she has in mind.

Determine how many months she could withdraw the stated monthly withdrawal in each of the following schemes.

Scheme 1 Put the money in a box under the bed and withdraw \$1500 each month to live on.

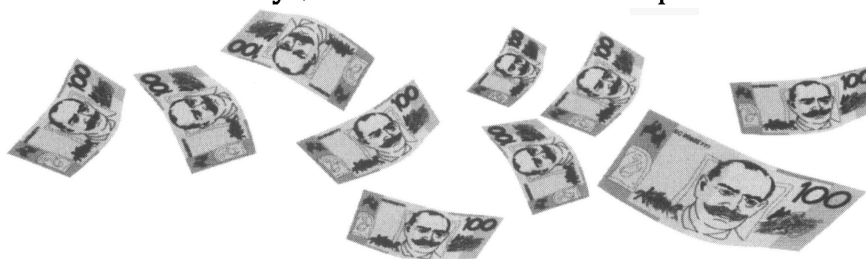
Scheme 2 Invest the entire \$120 000 in an account paying 8% interest per year, compounded monthly, and withdraw \$1 500 at the end of each month to live on.

Scheme 3 Same as scheme 2 but this time suppose the account pays interest of 9% per annum compounded monthly.

Scheme 4 Use \$20 000 of the \$120 000 to pay for flights and accommodation for an around the world holiday, invest the rest in an account earning 9% per annum interest, compounded monthly, and withdraw \$1500 each month during the holiday and thereafter.



Scheme 5 Invest the entire \$120 000 in an account paying 9% interest per year, compounded monthly. At the end of each of the first 12 months withdraw \$1500, at the end of each of the next 12 months withdraw \$1 600, at the end of each of the next 12 months withdraw \$1 700 and so on, increasing the monthly withdrawal by \$100 after each 12 month period.



Miscellaneous Exercise Four.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

- How much interest is earned in four years if \$4000 is invested in an account paying 12% per annum compounded monthly?
- A time series graph showing the number of people attending a particular music festival each year, for ten years, suggested that linear regression could be used to summarise the relationship between N , the numbers attending, and t the year number where $t = 1$ is used for the first year in the ten year period.
The least squares regression line for the data had equation:

$$N = 855t + 9459$$

Interpret the "855" in this equation.

- Giving each answer as a percentage and correct to three decimal places calculate the effective interest rate if the annual nominal rate is 6% and interest is compounded (a) quarterly, (b) monthly, (c) daily.
- To the nearest cent, what "one-off" lump-sum payment must be invested into an account paying compound interest of 8% per annum compounded quarterly for the investment to be worth \$5000 two years later?
- If we use the recursion relation $T_{n+1} = T_n \times 1.065$, $T_0 = \$4300$ the value of T_n gives the value of an account n years after a deposit was made to open the account, with interest compounded annually.
 - What was the amount of the deposit?
 - What was the annual interest rate?
 - Find the balance of the account after interest had been added at the end of the sixth year (to the nearest cent)?
 - What annual interest rate would achieve the balance that was the answer to part (c) but after just two years? (Give your answer as a percentage and correct to three decimal places).
- The table below relates to a savings account which Waylon opens by making an initial deposit and then adding a further fixed amount at the end of each month. Interest earned is compounded monthly with the same monthly interest rate throughout.

Month	Amount in account at beginning of month.	Amount in account at end of month, following adding of interest and deposit.
1	\$1600	\$1874.00
2	\$1874.00	\$2152.11

Determine the initial investment, the fixed monthly interest rate and the fixed monthly deposit.

7. (a) Determine the values that go in the spaces marked A, B, G.

t	Year	Months of Year	Sales	Four month mean for year	Sales $\div$ (four month mean for the year)
1	2010	First 4 months	442	395	1.1190
2		Second 4 months	299		A
3		Third 4 months	444		1.1241
4	2011	First 4 months	307	300	1.0233
5		Second 4 months	241		B
6		Third 4 months	352		1.1733
7	2012	First 4 months	294	261	1.1264
8		Second 4 months	203		C
9		Third 4 months	286		1.0958
10	2013	First 4 months	213	214	0.9953
11		Second 4 months	171		D
12		Third 4 months	258		1.2056
13	2014	First 4 months	170	E	1.0625
14		Second 4 months	122		F
15		Third 4 months	188		G

- (b) Create a completed version of the following tables.

Calculation of Seasonal Indices.						
	2010	2011	2012	2013	2014	Seasonal Index (3dp)
1st 4 Months	1.1190	1.0233	1.1264	0.9953	1.0625	1.065
2nd 4 Months						
3rd 4 Months	1.1241	1.1733	1.0958	1.2056		

Seasonally adjusted sales figures, S . (Nearest integer.)					
	2010	2011	2012	2013	2014
First 4 Months					
Second 4 Months					
Third 4 Months					

- (b) Use the ordered pairs (t, S) to determine the equation of the regression line $S = At + B$.
- (c) Use this regression equation and the seasonal indices to predict sales for the three 4 month intervals of 2015.